

Cellular and Medical Immunology

Adaptive Immune Response

Integrated Biomedical and Clinical Sciences OMFS513

Shymaa Bilasy, Pharm B., Ph.D.



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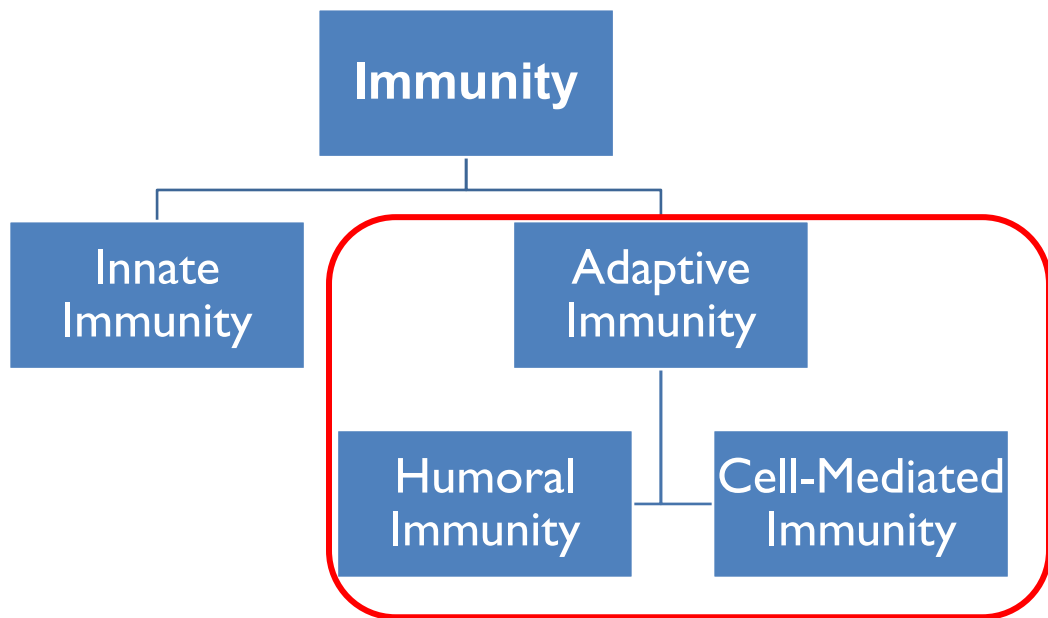
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Learning Objectives

1. Describe B and T-cell activation
2. Describe components of the adaptive immune response.
3. Describe the humoral immune response
4. Compare the different types of antibodies.
5. Describe the cell mediated immune response
6. Integrated immune response

Outline

- Lymphopoiesis (B and T-cell maturation and activation).
- Adaptive immune response:
 - Humoral
 - Cell mediated
- Integrated immune response against bacterial, viruses and fungal infections.



Features of Adaptive Immune System

- Diversity
 - B and T cells has diverse set of receptors to enable recognition of pathogens (VDJ arrangement)
- Self tolerant
 - B and T cell receptors should discriminate self from non-self antigens.
- Specificity
 - Receptors are specific for each antigen.
- Memory
 - Memory cells enable a faster and more efficient response on 2nd exposure.

Step 1: Lymphopoiesis and presence of functional Receptors

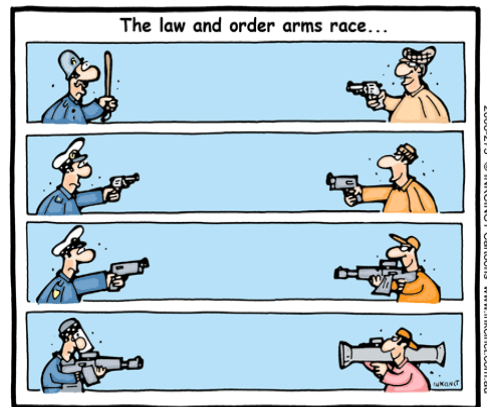
Goals:

- Generate diverse set of lymphocytes, each with unique antigen-specific receptor.
- Destroy self-reactive lymphocytes to prevent autoimmunity.
- Keep non-self-reactive lymphocytes and continue their maturation/activation in secondary lymphoid organs.

Antigen Specific Receptors: Generation of Diversity

Host-pathogen interactions = “arms race”

Generation of diversity is achieved via:
Antigen-specific receptors of individual B and T lymphocytes through recombination of
Variable (V) region gene segments with
diversity (D) and **joining** (J) gene segments.

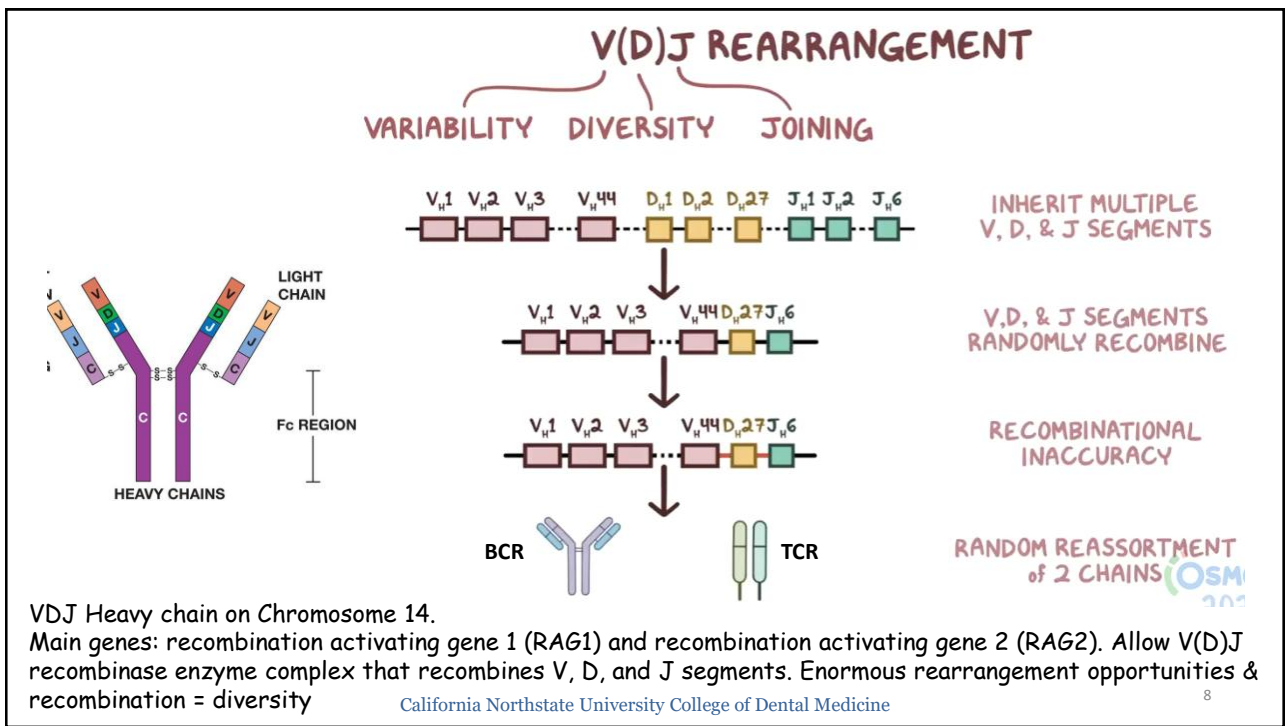


Mutators Enhance Adaptive Micro Evolution in Pathogenic Microbes

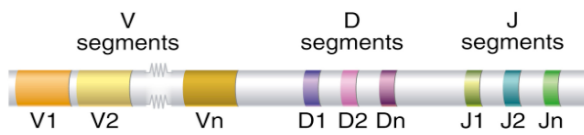
<https://www.researchgate.net/publication/358640910>

The Generation of Antibody Diversity <https://www.ncbi.nlm.nih.gov/books/NBK26860/>

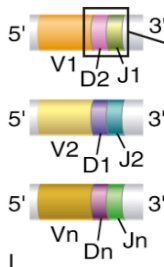
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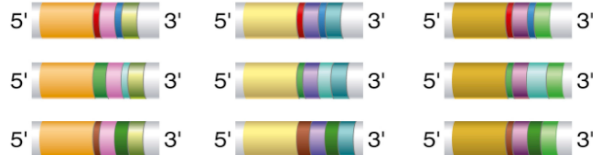
Germline DNA
(before somatic
rearrangement)



Combinatorial
diversity: Use
of different V, D,
and J segments



Junctional diversity:
Different N and P
nucleotides in
each clone



Millions of different
recombined antigen
receptor V genes

For Illustration

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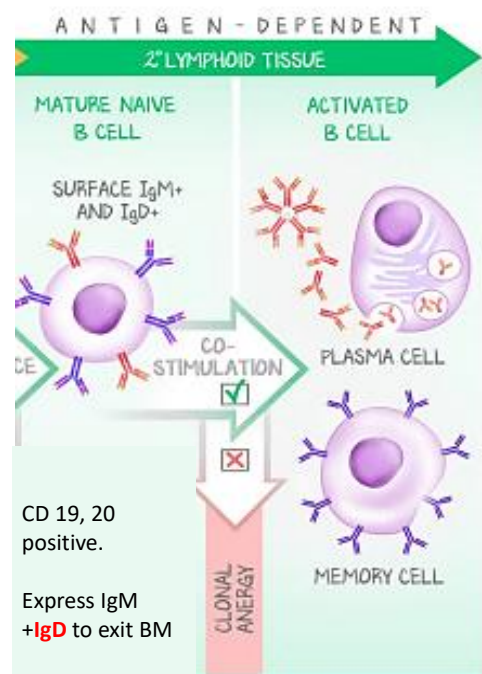
Development of B cells

A) Bone marrow:

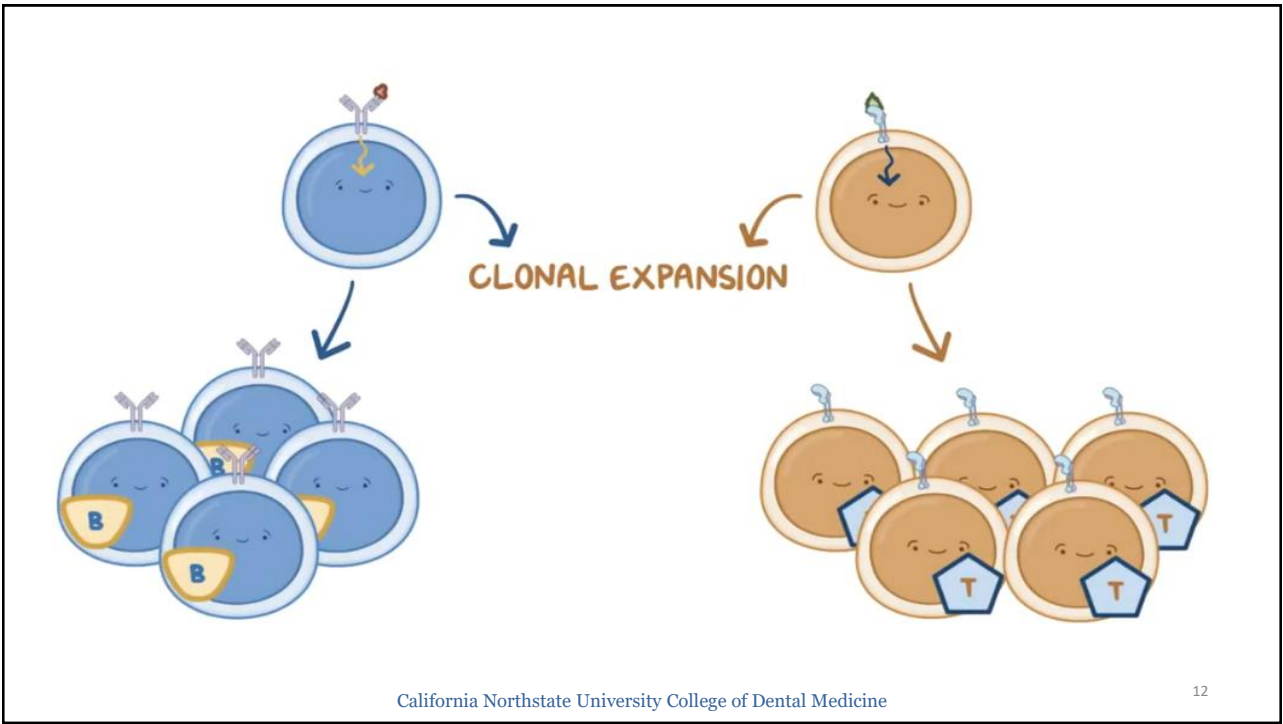
- Generally, IL-3, IL-4, IL-7 and B-cell activating factor (BAFF) are imp. for development.
- Six stages from common lymphoid progenitor cells to pro-B cells,..... Immature B cells. During these stages, B cell is building the BCR and Rag-1 and Rag-2, are important for B cell development and VDJ arrangement.
- Following the construction of IgM receptor, B cells are exposed to self-antigens.
 - Positive selection: for functionality
 - Negative selection: test self vs non-self recognition.
- Mature B cells has a B-cell receptor = membrane-bound immunoglobulin molecule. Each B cell clone will have its unique receptor (IgM).

Next, cell starts expressing IgD along with IgM (ticket out of bone marrow).

In secondary lymphoid organs (Ag-dependent), B cells are exposed to antigens. If, activated B cells rapidly multiply via clonal expansion to generate identical daughter cells.



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Step 2 : Antigen presentation and activation

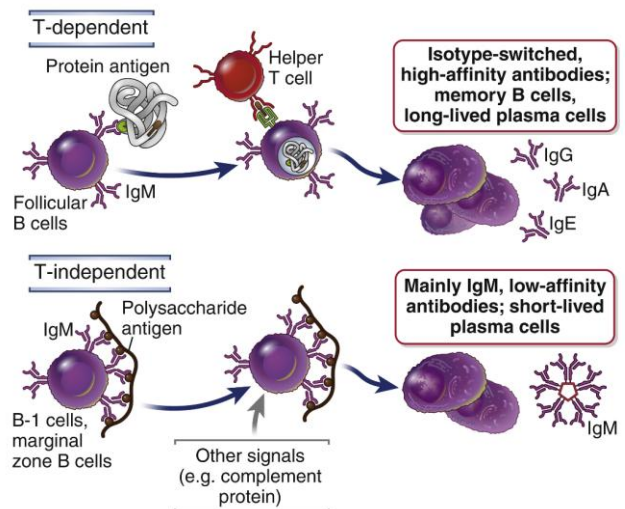
Humoral Immune Response

Activation

For full activation, B cells need a two-step mechanism:

- 1st they need to recognize an antigen
- 2nd receive a co-stimulatory signal

- T-cell dependent: Receiving the co-stimulation from helper T cell (TCR identifies the antigen presented on the B cell via MHCII). Mostly follicular B-cells.
- T-cell independent: co-stimulation via danger signal from IFN γ . Mostly B-1 and marginal zone B cells.



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Subsets of B cells

1- From bone marrow HSCs

Follicular B cells (B-2 B cells): Major population. Where? ----- . Express IgD and IgM and can respond to stimulation. Most mature B cells belong to the follicular B cell. High diversity.

Activated B cells differentiate into:

1- Effector cells “plasma cells”.

Plasma cells change CD markers expression & drop surface immunoglobulin. Instead, it makes secreted antibodies. Those activated B cells produce IgM before “class switching”.

2- Memory cells: Long-lived B cells that promote an adaptive immune response to a pathogen upon subsequent exposure.

Marginal zone B cells (Where?): express IgM without IgD. Life span>follicular B cells with self renewal. Limited diversity, differentiate into short-lived IgM-secreting plasma cells.

Function: -----? and keep the organization of the marginal zone.

2- Fetal liver-derived

B-1 B cells: Found as a self-renewing population in the peritoneum and mucosal sites. Limited diversity. Can produce “natural antibodies” against commensal microbiota, and ABO blood groups (IgM). Also, around 50 % of mucosal IgA can be derived from B-1 cells.

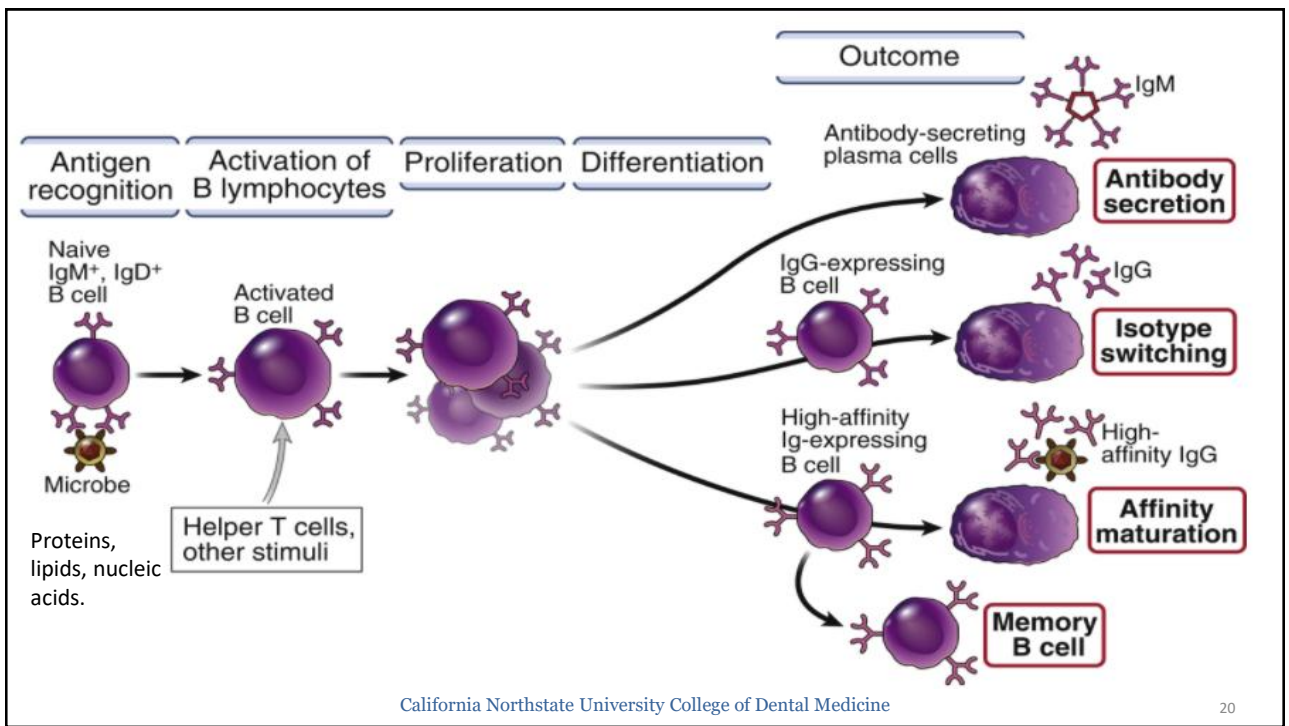
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Step 3: Processes of adaptive immune response

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1- Antibodies

Antibodies are circulating glycoproteins that are produced in response to exposure to foreign particulates.

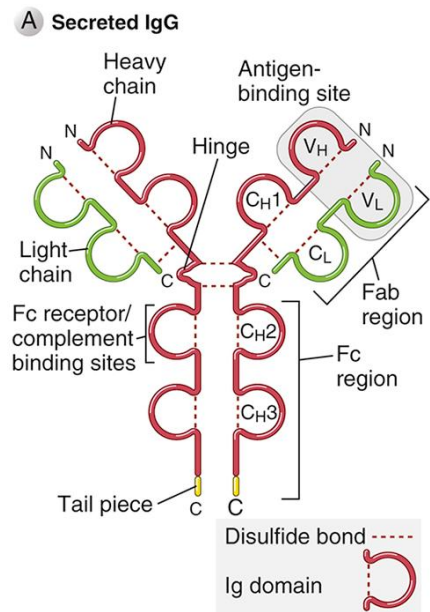
Antibodies, aka immunoglobulins (Ig; immunity-conferring portion of serum globulins) or gamma globulin.

Immunoglobulins are synthesized only by B lymphocytes and exist in two forms:

- 1. Membrane-bound Igs on the surface of B lymphocytes (antigen receptors).**
- 2. Secreted antibodies by plasma cells (opsonization and immune response). Plasma cells can secrete 2000 antibody/sec**

Antibody Structure

- Two identical **light chains** [(lambda (λ)) and kappa (κ)], and two identical **heavy chains** joined by disulfide bonds.
- Five types of heavy chains α , δ , ϵ , γ and μ for IgA, IgD, IgE, IgG, and IgM antibodies, respectively.
- Variable region at the amino terminal and a constant region at the carboxy-terminal.
- Antibodies display remarkable variability in the variable region that bind antigens.



Courtesy of Dr. Alex McPherson, University of California, Irvine.

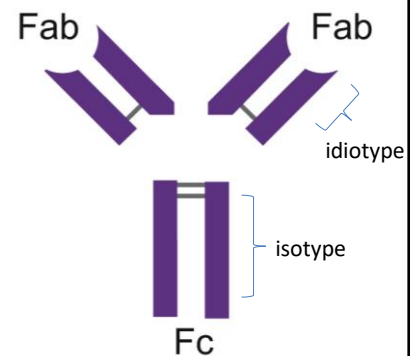
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Antibody Fragments

Proteolytic cleavage by papain yield:

1- Two Fab (fragment antigen binding). This fragment contains the antigen binding site or paratope.

2- One Fc (fragment crystallizable) fragment. This fragment mediate the antibody effector function (like complement activation, opsonization, neutralization of microbe) and can be bound to the B-cell receptor.



Valency, Affinity and Avidity

Valency = number of Fab fragments.

Upon encountering the specific antigen, the antibody binds this antigen with its variable region via variety of bonds (noncovalent bonds, or hydrogen bonds, or electrostatic bonds, or hydrophobic interactions).

Affinity = The strength of the binding between a single binding site of an antibody and an epitope of an antigen.

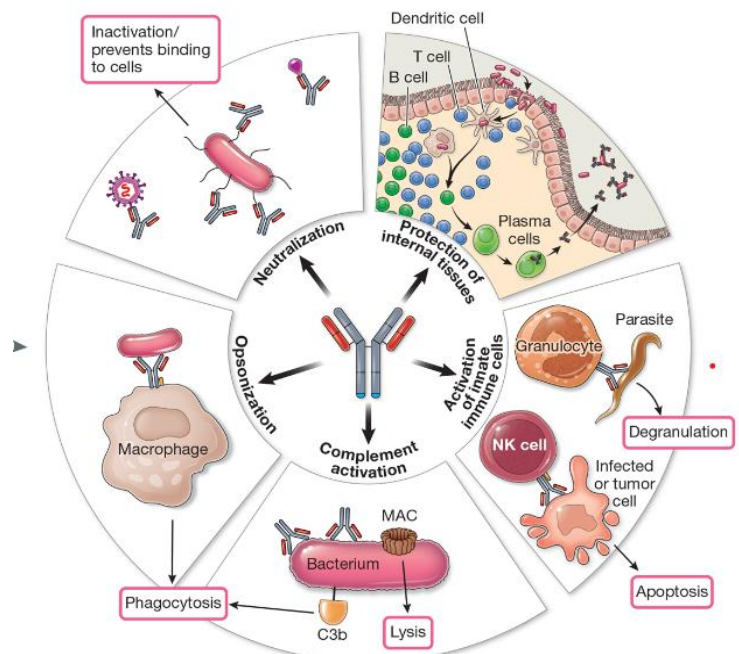
Avidity = The overall strength of attachment of an antibody unit to an antigen. IgG avidity test can differentiate recent from past/chronic infections (CMV infection).

Cross reactivity = antibodies that can recognize (or has high affinity) epitopes of different antigens.

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Classes of Antibodies

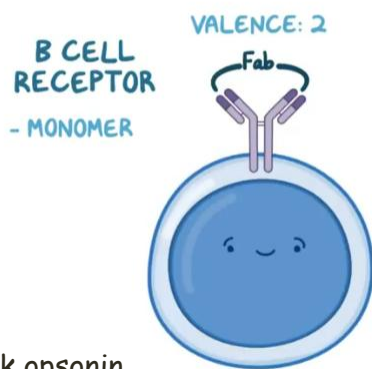


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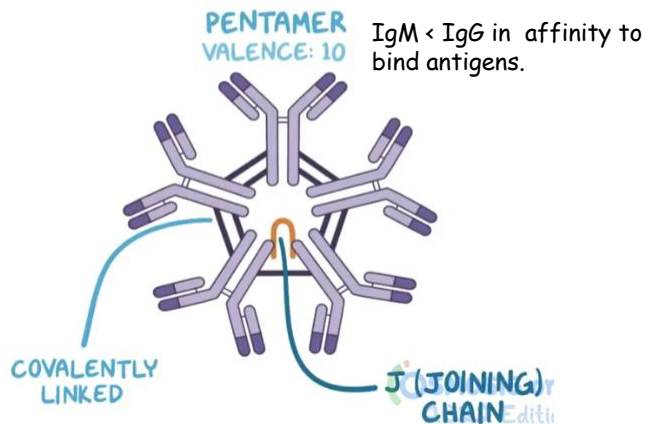
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IgM IMMUNOGLOBULIN M

- * 1st ANTIBODY RESPONSE MADE
- * WORKS AGAINST CARBOHYDRATE & LIPID ANTIGENS



- * 4% of IMMUNOGLOBULIN in SERUM
- * MOST EFFECTIVE at ACTIVATING COMPLEMENT PATHWAY



- Weak opsonin
- Cannot cross placenta
- IgM can be activated with/without T-cell influence, which happens upon the activation of B cell receptors by bacterial carbohydrates/lipids.

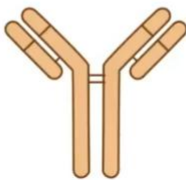
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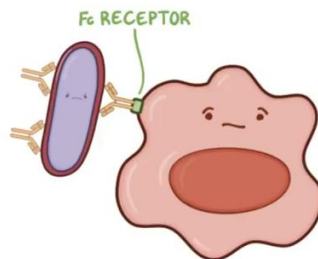
IgG

* MOST ABUNDANT IMMUNOGLOBULIN

MONOMER
VALENCE: 2



* SERVES as an OPSONIN
~ HELP PHAGOCYTES GRIP BACTERIA

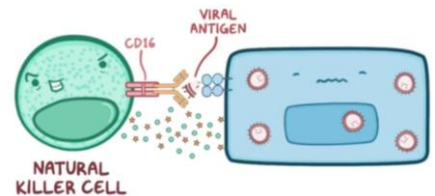


70%-75% of serum antibody pool.
IgG has the longest half life among all Igs (around 3 weeks)

* ACTIVATES the CLASSICAL COMPLEMENT PATHWAY

ANTIBODY DEPENDENT CELL MEDIATED CYTOTOXICITY (ADCC)

For intercellular pathogens, Work with NK cell to perform cell mediated toxicity



Cross the placenta

Four subclasses of IgG

- **IgG1:** most abundant subclass. Major contributor to response for protein antigens (viral proteins, toxoids).
- **IgG2:** 20 % of IgG population. Imp for response to polysaccharide antigens (bacterial capsules).
- **IgG3:** around 8 % of IgG population. Shortest half-life (7 days). Strongest complement activation.
- **IgG4:** least abundant. Produced after prolonged or repeated antigen exposure.

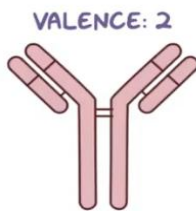
Class switching goes from IgG3 to IgG1 to IgG2 to IgG4. System may aim to down-regulate inflammation in chronic antigen exposure (IgG2 and IgG4).

Main functions of IgG:

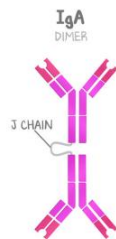
- 1- Opsonization of antigens.
- 2- Activates complement system via classical pathway.
- 3- ADCC (antibody-dependent cell mediated cytotoxicity). Binding of IgG to viral antigens will activate NK cells (via CD16) and induce the release of granzymes and perforin to kill the target cell.
- 4- Neonatal immunity. Only antibody to cross the placenta
- 5- Passive immunity.

IgA Two forms IgA1 and IgA2

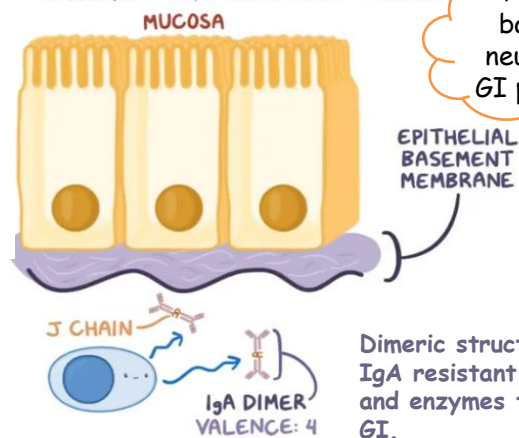
- 20% SERUM IMMUNOGLOBULIN
- EXISTS as a MONOMER



- OPSONIN for PHAGOCYTES like
~ EOSINOPHILS, NEUTROPHILS,
& some MACROPHAGES



- MAIN IMMUNOGLOBULIN in MUCOSAL SITES like
~ SALIVA, TEARS, BREAST MILK & SEMEN



Protects babies by neutralizing GI pathogens

Dimeric structure makes IgA resistant to acids and enzymes found in the GI.

Monomer in plasma, dimer in secretions.
IgA is secreted in saliva, tears, and colostrum.

IgA functions/properties

- IgA is found in high concentrations in secretions.
- Half-life of the dimer around 5 days.
- IgA provide protection via:
 - coat pathogens and prevent bacterial colonization.
 - Preventing formation of dental plaque.
 - Inhibit viral adherence
 - Block the activity of biologic toxins
- In periodontitis, IgG (in gingival crevicular fluid) and IgA (in saliva) are important for pathogenesis.
- Microbial adaptation: IgA protease is secreted by certain bacteria like *S. pneumoniae*, *Neisseria* and *H. influenza* to cleave IgA and allow mucosal colonization.

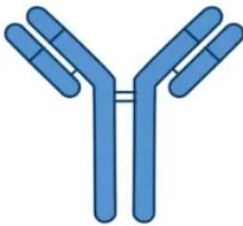
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IgE

* ASSOCIATED with ALLERGIC & ANTI-PARASITIC RESPONSES

MONOMER
VALENCE: 2



* 0.004% TOTAL SERUM IMMUNOGLOBULIN

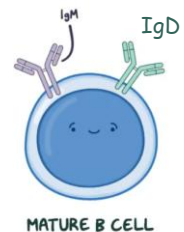
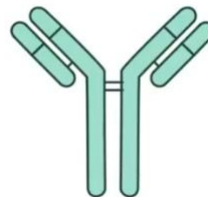
IgD

* LESS THAN 1% of SERUM IMMUNOGLOBULIN

Signal the B-cell to leave the bone marrow

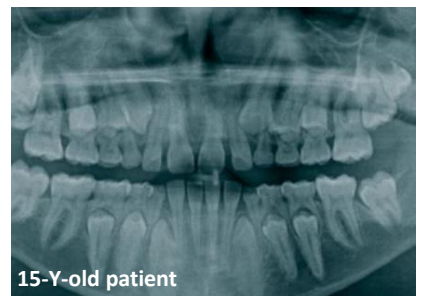
Enhance mucosal homeostasis

MONOMER
VALENCE: 2



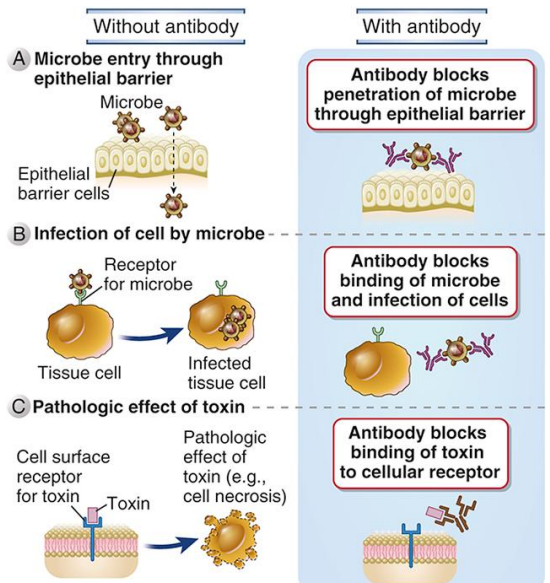
Hyper-IgE syndrome (Job's syndrome)

- Due to genetic mutation in *STAT3* (autosomal dominant).
- Characterized by high levels of serum IgE.
- Clinical manifestations
 - Eczema
 - Recurrent pulmonary infections.
 - Increased risk of fractures
 - Dental anomalies:
 - retention of deciduous teeth.
 - ectopic eruption or non-eruption of permanent teeth.
 - Increased susceptibility to caries



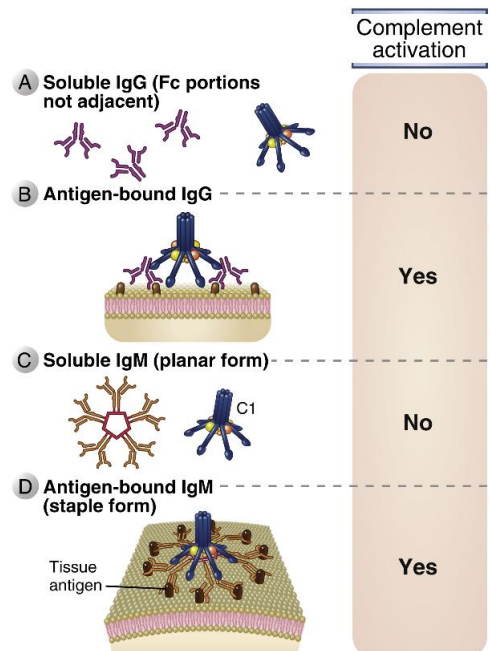
Functions of Antibodies

- 1- Neutralization of microbes and microbial toxins.
- 2- Antibody-mediated opsonization and phagocytosis. *IgG functions as an opsonin that coat microbes. IgG "FC" region is recognized by phagocytes and activates phagocytosis.*
- 3- Antibody-Dependent Cell-Mediated Cytotoxicity. *NK cells bind to antibody-coated cells and secrete cytokines to mediate cytotoxicity.*

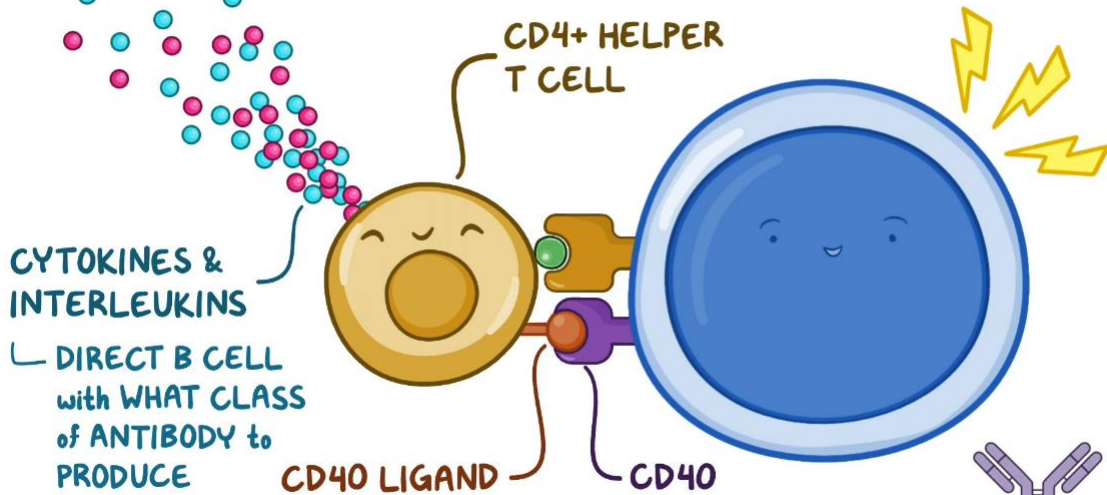


4- Mediate complement activation via the Fc region (*mainly IgM and IgG antibodies bound to antigens*)

5-Mediate maternal-neonatal Immunity. IgG, across the placenta, while maternal IgA, in breast milk.



2- Isotype Switching



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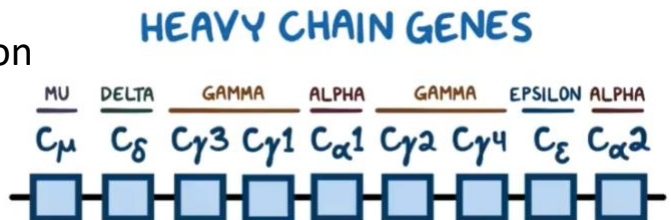
iOSMosis.org

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- Requires CD40-CD40L interaction.
- During isotype (class) switching:
 - Fab fragments are maintained
 - parts of the DNA encoding the FC regions are deleted by alternative splicing.

- Depends on cytokine stimulation

- IL-4 and IL-5 stimulate IgE
- IFN γ stimulate IgG.
- TGF β stimulate IgA



- Once switched, they don't go back to IgM.

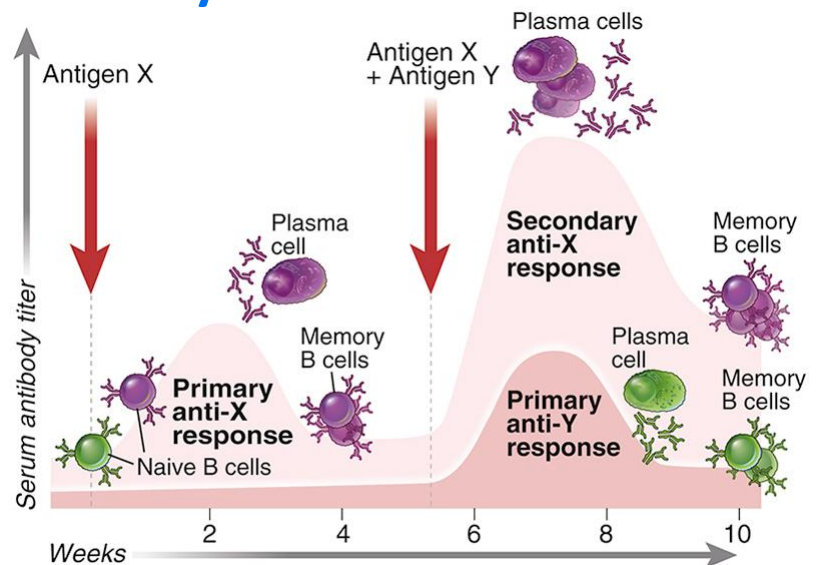
3- Affinity maturation & Somatic Hypermutation

BCR are fine-tuned !

- Somatic hypermutation is process where B-cell can change the BCR receptors to increase their affinity for an antigen.
- Following activation, the VDJ regions undergo high-rate changes named “somatic hypermutations” to enhance in the antigen recognition in the paratope.
- B cell clones with the highest affinity to the antigen will have higher proliferation rate “affinity maturation”.

4- Memory B cells

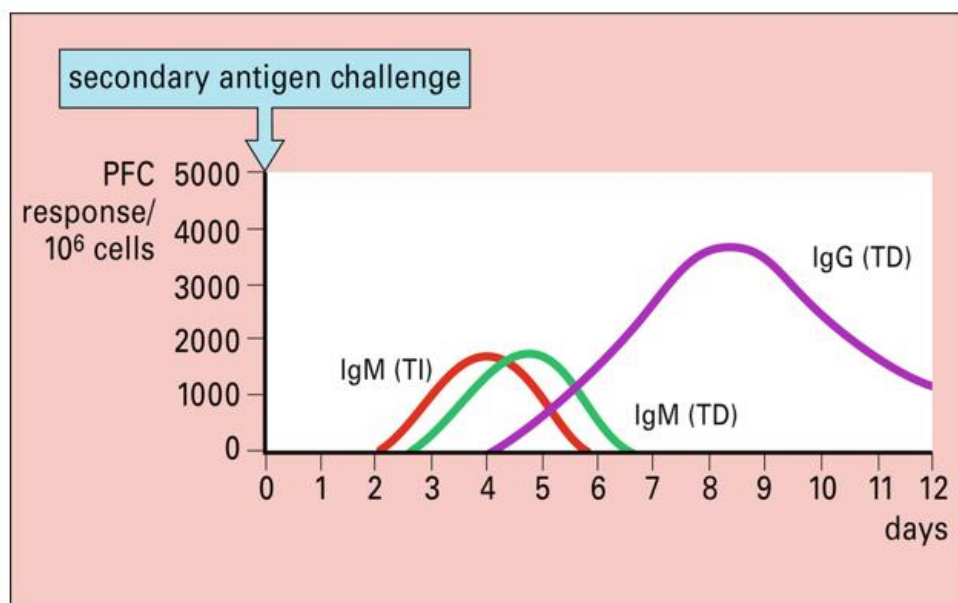
In the germinal center, B cells differentiate to memory B cells. These cells are class-switched and they provide an amplified and quicker immune response during the second exposure.



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Immune response to T-dependent (TD) and T-independent (TI) antigens *in vitro*.

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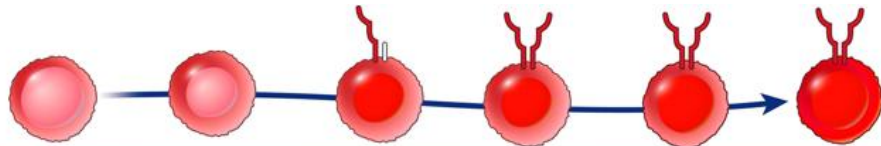
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Cell-Mediated Immune Response

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Step 1: T-cell Development, Maturation

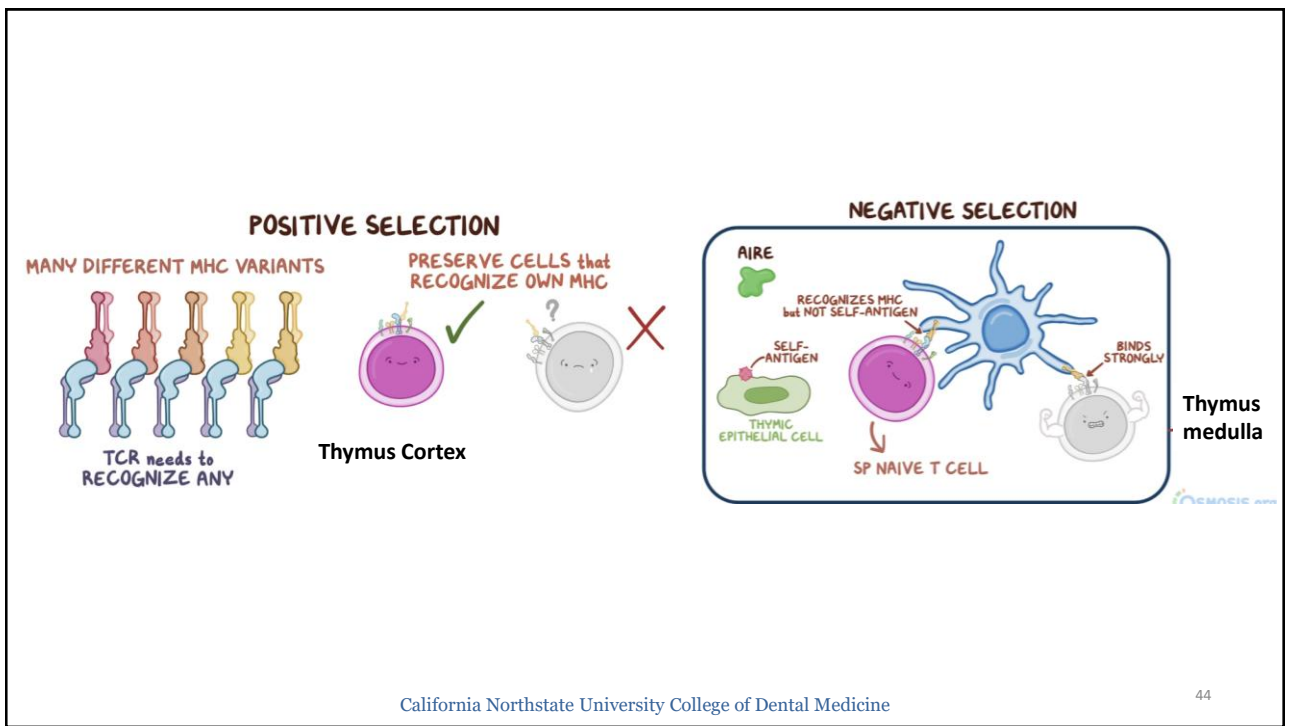


Stage of maturation	Stem cell	Negative		Double positive CD4 and CD8 +ve	Single positive (immature)	Mature T cell
Anatomic site	Bone marrow	Thymus			CD4 Or CD8 +ve	Periphery
Response to antigen	None	None	None	Positive and negative selection		Activation (proliferation and differentiation)
		TCR receptor arrangement				

- All T cells express the CD3 molecule, which is part of the T cell receptor.
- Pro-T cell are CD negative. As the cell mature, it develop CD3 in addition to CD4 and CD8 (double positive stage).
- Finally, they become single either positive CD4 or CD8 cells.

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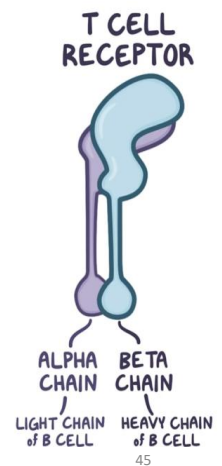
Step 2: Activation

Cell-Mediated Immune Response

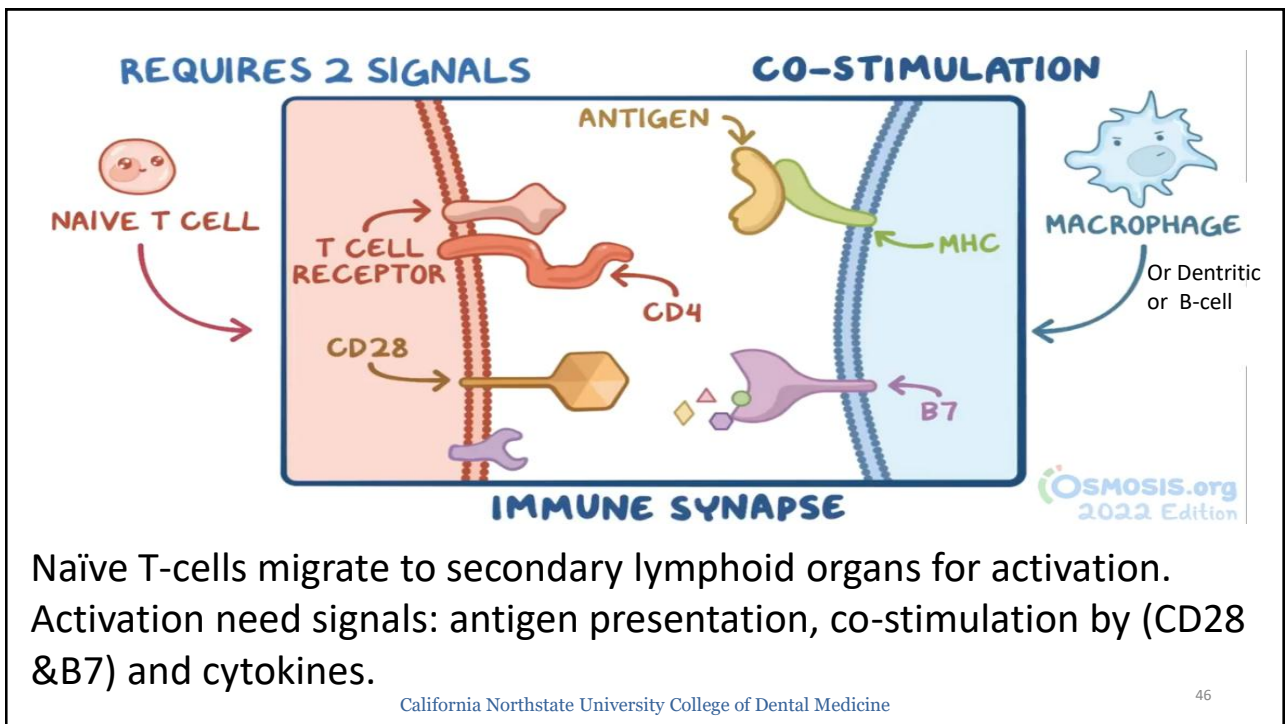
Naive T-lymphocytes recognize antigens of cell-associated microbes (APCs like dendritic cells, B-cells, macrophages) presented on host major histocompatibility complex (MHC, HLA).

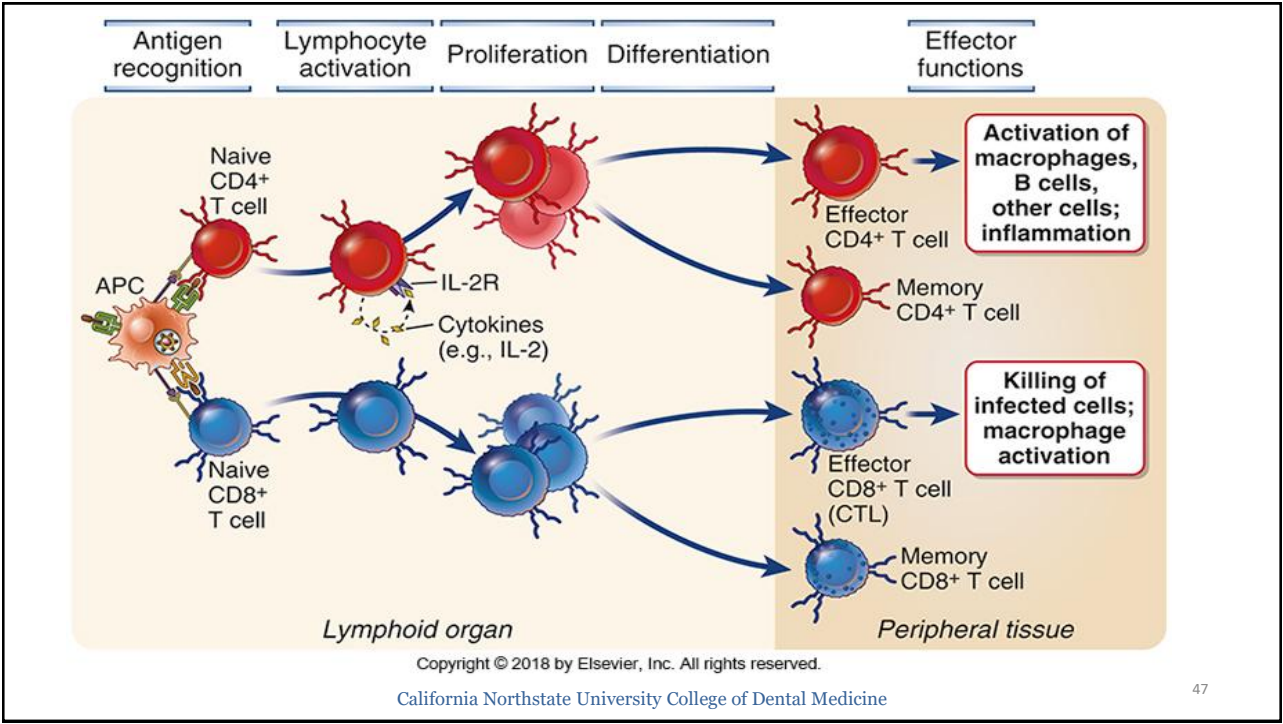
- Recognize class I MHC become cytotoxic T cells (CD8 positive). Class I MHC are expressed on all nucleated cells.
- Recognize class II MHC become T-helper cells (CD4 positive). Class II MHC are expressed on dendritic cells, macrophages, and B lymphocytes.

IL-2, IL-7 and IL-17 are important for T cell development & maturation.



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Subsets of Effector T helper Cells

T helper
cell 1

T helper
Cell 2

T helper
cell 9

T helper
Cell 17

T helper
cell 22

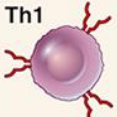
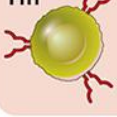
Follicular T
cells

Functions

- **Th1 cells** activate macrophages to ingest and destroy microbes via the secretion of IFN- γ .
 - IFN- γ is the principal macrophage-activating cytokine (phagocytosis). Promotes the differentiation of CD4+ T cells to the Th1. Inhibits Th2 and Th17 development.
 - IFN- γ activates macrophages and upregulate cytokines production from macrophages, DCs, and others (positive feedback).
 - Th1 secrete IL-2 to activate cytotoxic T cells.
- **Th2 cells** stimulate IgE-, mast cell–. and eosinophil-mediated reactions to eradicate helminthic infections and promote tissue repair.
 - IL-5, activates eosinophils and mast cells. IL-4 cause IgE class switching. IL-13 increase mucus secretion and diverse actions

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- **Th17 cells** combat fungal and extracellular bacteria by recruiting leukocytes (mainly, neutrophils), to sites of infection (via IL-17 & 22).
- **T Follicular Helper (Tfh)** cells facilitate humoral immune response.
 - Through IL-21 production, Tfh cells promote the differentiation of B cells into Ig-producing plasma cells.
 - Play role in class switching and memory B-cell development in the germinal zone.
- **Th9 cells** (produce IL-9).
 - Abundant in the skin and mucosal tissues. Involved in mucosal immunity.
 - Th9 cells have been implicated in allergic diseases, including asthma and in defense against parasites, and other.

Effector T cells	Defining cytokines	Principal target cells	Major immune reactions	Host defense	Role in disease
Th1 	IFN- γ	Macrophages 	Macrophage activation	Intracellular pathogens	Autoimmunity; chronic inflammation
Th2 	IL-4 IL-5 IL-13	Eosinophils 	Eosinophil and mast cell activation; alternative macrophage activation	Helminths	Allergy
Th17 	IL-17 IL-22	Neutrophils 	Neutrophil recruitment and activation	Extracellular bacteria and fungi	Autoimmunity; inflammation
Tfh 	IL-21 (and IFN- γ or IL-4)	B cells 	Antibody production	Extracellular pathogens	Autoimmunity (autoantibodies)

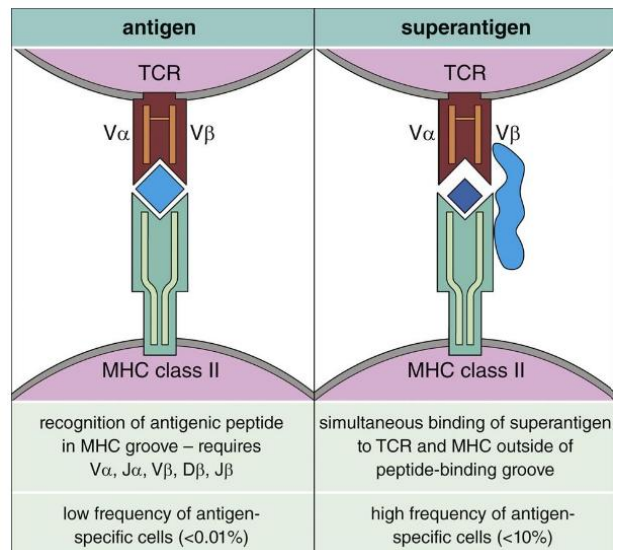
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Superantigens

- Example: staph. & strept. superantigens.
- Trigger stronger immune responses and T-cell proliferation leading to massive release of pro-inflammatory cytokines (IL-2, TNF α and TNF β as well as IL-1 β from activated macrophages)
- Contribute to microbial pathogenicity



Cytotoxic T cells

Helper T cells promote the activation of the cytotoxic T cells.

Functions of cytotoxic T cells:

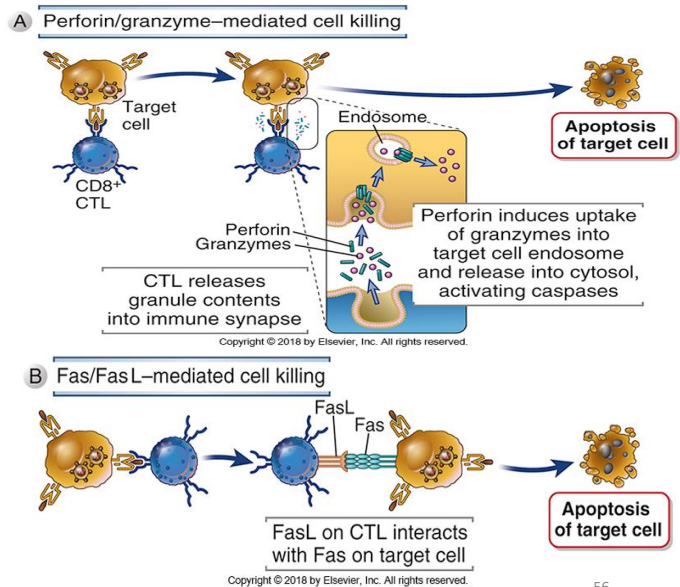
- 1) Eliminate intracellular pathogens. This includes viruses and certain bacterial species that can reside in the cytosol of infected cells or macrophages (to evade phagosomal digestion).
- 2) Mediate anti-tumor immunity.

Mechanism of Cytotoxic T cell action

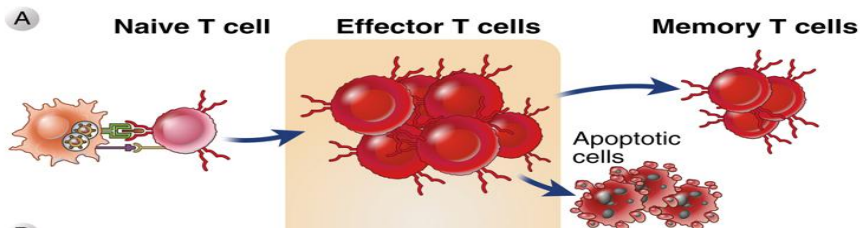
1) Triggering apoptosis

- a) Granule exocytosis (perforin and granzyme).
- b) Fas (FS-7-associated surface antigen)/FASL pathway via activating caspase cascade and apoptosis.

2) Secretion of cytotoxic cytokines like $\text{TNF}\alpha$ and induce apoptosis.



Memory T cells



- Memory cells express increased levels of antiapoptotic proteins (may explain the prolonged survival).
- Memory cells can migrate to peripheral tissues and respond to antigens at the site of infection. Whereas, naive T cells migrate to secondary lymphoid organs.

Adaptive Immunity in Oral Cavity

- T and B cells respond with high specificity and sustained protection (1-2 % of total leukocyte population).
- CD4⁺ T cells (Th1, Th2, and Treg cells) are more prominent found in oral cavity.
 - Th1 cells are involved in the bone loss characterized in chronic periodontitis by stimulating osteoclasts.
 - Th2 and Treg cells mainly regulation of the immune response (incl. inhibition of bone loss in periodontitis).
- B cell: humoral response through IgA secretion.
 - In chronic periodontitis, the saliva levels of the B cell activating factor (BAFF) is elevated compared to healthy control

The two arms of the adaptive immunity

	Humoral immunity	Cell-mediated immunity
Microbe	Extracellular microbes	Extracellular microbes Phagocytosed microbes that can live within macrophages Intracellular microbes (e.g., viruses) replicating within infected cell
Responding lymphocytes	B lymphocyte	Helper T lymphocyte Cytotoxic T lymphocyte
Effector mechanism	Secreted antibody	Activated macrophage Neutrophil Killed infected cell
Functions	Block infections and eliminate extracellular microbes	Activated phagocytes kill microbes Kill infected cells and eliminate reservoirs of infection

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HBV Diagnosis

A) Direct detection of viral DNA (viral load).

B) Antigens

- Hepatitis B surface antigen (HBsAg)
- Hepatitis B envelope antigen (HBeAg)

C) Antibodies

- Anti-hepatitis B surface antigen (Anti-HBsAg)
- Anti-hepatitis B envelope antigen (Anti-HBeAg)
- Anti-hepatitis B core antigen (Anti-HBcAg).

D) Elevated ALT/AST levels

- In the acute phase, you can detect the viral DNA, HBs antigen (appear within 1-2 weeks post infection) or HBe antigen (appear shortly after but clears quickly) and detect the IgM antibodies. In general, antibodies titer increase as acute infection resolves.
- In the chronic phase, anti HB core IgG and HBs antigen are present for >6 month. Anti HB core IgM is negative.
- Recovery “Window phase”: Anti HB core IgG antibodies
- Vaccinated: anti HBs IgG

DIAGNOSIS: ACUTE HEP B

↳ ANTIGENS & ANTIBODIES

anti HB core IgM
ANTIBODIES



HBs



HB_e



VIRAL DNA



HB core IgG
ANTIBODIES
↳ DEPENDS on TIME
of TESTING

DIAGNOSIS: CHRONIC HEP B

anti HB core IgM
ANTIBODIES



-

anti HB core IgG
ANTIBODIES



> 6 MOS.

HB_s ANTIGEN



> 6 MOS.

VIRAL DNA



+/-

HB_e ANTIGEN



+/-

HB_e ANTIBODIES



+/-

RESOLVED HEPATITIS

anti HB core IgG
ANTIBODIES



+

anti HB_s
ANTIBODIES



+/-

anti HB_e
ANTIBODIES



+/-

ANTIGENS



-

anti HB core IgM
ANTIBODIES



-

VIRAL DNA



-

OSMOSIS.01

HEPATITIS B SEROLOGIC MARKERS

	VACCINATION	ACUTE HEPATITIS	RESOLVED ACUTE HEPATITIS	CHRONIC HEPATITIS
HBsAg	-	+	-	+
Anti-HBs	+	-	+	-
IgM anti-HBc	-	+	-	-
IgG anti-HBc	-	+	+	+
HBeAg	-	+	-	+/-
Anti-HBe	-	+	+	+

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References

